ARUNACHALAM S

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SUMMARY

Mechanical Engineer with a B.Tech from NIT Trichy and experience in manufacturing and process improvement projects at Senvion, a leading wind turbine manufacturer. Skilled in project coordination, process optimization, and continuous improvement, with hands-on expertise in FEA tools (ANSYS, SpaceClaim). Proven ability to manage timelines, collaborate across teams, and implement solutions that enhance efficiency and quality. Passionate about applying technical and project management skills to deliver successful outcomes in dynamic engineering environments.

SKILLS

- Process Optimization
- Lean Manufacturing
- Project Management
- Ansys Workbench
- AutoCAD
- SolidWorks
- Ansys SpaceClaim
- Power BI
- Microsoft Office

WORK EXPERIENCE

Manufacturing Excellence Engineer at Senvion Wind Technology Pvt Ltd — **Jun 2024- Present**

- Achieved a 71% reduction in curing cycle time by leading a Curing Optimization Project, aligning processes with industry standards and driving efficiency gains.
- Leveraged Power BI for data-driven cycle time analysis, implementing step-by-step corrective actions to optimize production and reduce lead times.
- Initiated and executed a Paste Consumption Optimization Project, reducing material costs while maintaining product quality.
- Conducted comprehensive Root Cause Analysis (RCA) to minimize product defects, directly supporting the organization’s ZERO Defect initiative.
- Designed and implemented kitting trolleys using AutoCAD, streamlining material handling and improving shop floor logistics efficiency by 30%.
- Performed advanced Finite Element Analysis (FEA) of wind turbine tower components using ANSYS Workbench and SpaceClaim, preparing detailed technical reports to support design and process decisions.

Research Assistant at Bureau of Indian Standard, Delhi — **May 2023 - Jul 2023**

- Contributed to the development of an Indian standard for pneumatic door closure mechanisms in buses, focusing on safety, reliability, and performance criteria.
- Conducted a comparative study of door closing systems through technical literature & industry resources.
- Performed field and industrial visits to Ashok Leyland, Delhi Transport Corporation, and International Centre for Automotive Technology (ICAT) to validate design practices and collect technical data.
- Consolidated research findings, technical inputs, and compliance parameters to prepare a working draft for the standardization process.

PROJECTS

Thermo-Mechanical Analysis of Carbon Fiber (CF) Reinforced PLA

- Prepared standard samples of CF-reinforced PLA and neat PLA using a MakerBot Replicator 3D printer for mechanical and thermal testing.
- Conducted compression, tensile, and thermal conductivity tests on 4 infill patterns (diamond, cat, Hilbert, hexagonal) and 3 infill densities (25%, 50%, 75%) to evaluate performance variations.
- Analyzed results to compare tensile strength, compression strength, and thermal conductivity, finding higher thermal conductivity and higher compression strength in CF-reinforced PLA, while neat PLA showed superior tensile strength.
- Gained hands-on experience in additive manufacturing, composite materials testing, and data analysis.

Effect of Nanoparticles on CI Engine Characteristics

- Operated test setups and equipment to measure performance, emissions, combustion, noise, and vibration of a CI engine under varying fuel conditions.
- Studied the impact of 12 biodiesel blends with 3 different concentrations of cerium-oxide nanoparticles on engine characteristics.
- Supported thesis work by preparing schematic diagrams, graphs, and data analysis for technical reporting.
- Gained hands-on experience in engine testing, fuel analysis, and experimental data interpretation.

Hybrid Battery Thermal Management using Nanofluids for EVs

- Designed and simulated a novel hybrid battery thermal management system combining air and liquid cooling.
- Tested the system with multiple battery cell configurations using three liquid coolants, including Cu-nanofluids and CNT-nanofluids, to identify the optimal topology.
- Achieved a configuration that limited maximum battery temperature to 308.078 K, enhancing thermal performance and safety.
- Gained hands-on experience in thermal simulation, battery cooling strategies, and performance optimization

EDUCATION

B.Tech in Mechanical Engineering — **Jun 2020 - May 2024**
National Institute of Technology, Tiruchirappalli — *CGPA: 8.51*

Class 12th — **May 2020**
The Velammal International School, Ponneri — *Percentage: 90.8%*

Class 10th — **May 2018**
Pon Vidhyashram, Kolapakkam, Chennai — *Percentage: 83.8%*

ADDITIONAL INFORMATION

- **Languages:** Tamil, English & Hindi
- **Certifications:** MTA: Introduction to Programming using Python (Microsoft)
- **Awards/Activities:**

1. I have secured an All India Rank 582 in GATE 2026 (AE).
2. Won 3rd Prize in Quality Slogan Competition conducted by SWTPL.